The Iambic-Trochaic Law without Iambs or Trochees: Parsing speech for grouping and prominence

Esmail Moghiseh, Morgan Sonderegger, and Michael Wagner

Department of Linguistics and Centre for Brain, Language, and Music, McGill University, Montréal, QC H3A 1A7, Canada

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Listeners parse the speech signal effortlessly into words and phrases, but many questions remain about how. One classic idea is that rhythm-related auditory principles play a role, in particular that a psycho-acoustic 'Iambic-Trochaic Law' (ITL) ensures that alternating sounds varying in intensity are perceived as recurrent binary groups with initial prominence (trochees), while alternating sounds varying in duration are perceived as binary groups with final prominence (iambs). We test the hypothesis that the ITL is in fact an indirect consequence of the parsing of speech along two in-principle orthogonal dimensions: prominence and grouping. Results from several perception experiments show that the two dimensions, prominence and grouping, are each reliably cued by both intensity and duration, while foot type is not associated with consistent cues. The ITL emerges only when one manipulates either intensity or duration in an extreme way. Overall, the results suggest that foot perception is derivative of the cognitively more basic decisions of grouping and prominence, and the notions of trochee and iamb may not play any direct role in speech parsing. A task manipulation furthermore gives new insight into how these decisions mutually inform each other.

I. INTRODUCTION

Bolton (1894, 232), in his classic study on rhythmic perception, observed a regularity in the perception of tone sequences: "The general principle is this: In a series of auditory impressions, any regularly recurrent impression which is different from the rest, subordinates the other impressions to it in such a way that they fall together in groups. If the recurrent difference is one of intensity, the strongest impression comes first in the group and the weaker ones after. If the recurrent difference is one of duration, the longest impression comes last." In other words, English listeners perceive a series of trochees if every other sound is louder, and a sequence of iambs if every other sound is longer, a finding confirmed in Woodrow (1909) and many studies since (Bell, 1977; Bhatara *et al.*, 2013; Fraisse, 1956; Hay and Diehl, 2007; Iversen *et al.*, 2008; Rice, 1992; Vos, 1977, i.a.).

This pattern is only observed for sounds within a certain time range. Bolton (1894, 214), for example, notes that in his experiment, tone sequences were not perceived as structured into groups when separated by less than 115ms or when separated by more than 1.58s, but these values depend on factors such as the length of the individual tones (Fraisse, 1956, i.a.). Crucially, this time range subsumes the rate at which syllables and words occur in natural speech. This has prompted the hypothesis that Bolton's principle reveals a general auditory principle, which today is often referred to as the 'Iambic-Trochaic Law' (ITL) following Hayes (1995), which has been accorded a central role in how humans parse and cognitively represent the speech signal (Bion *et al.*, 2011; Hay and Diehl, 2007; Hayes, 1984, 1995; Nespor *et al.*, 2008; Peña *et al.*, 2011).

Hay and Diehl (2007), for example, hypothesized that the Iambic-Trochaic Law reflects a domain-general auditory bias, which can guide the parsing of the speech signal before a child has learned the words of the language. They further speculated that languages with iambic word stress would tend to use duration as a cue, while languages with trochaic stress would use intensity. Similar ideas about distribution of cues encoding foot type at the word and phrasal level are discussed in Iversen *et al.* (2008); Patel (2007) and Nespor *et al.* (2008). A language-independent parsing strategy based on acoustic cues along these lines could help children bootstrap the signal during language acquisition (Bion *et al.*, 2011; Hay and Saffran, 2012; Yoshida *et al.*, 2010), and could complement strategies that are based on distributional regularities within the language itself, such as transitional probabilities (Aslin *et al.*, 1998); see Mattys and Bortfeld (2016) for a review.

There are conflicting results, however, with respect to how consistently the ITL applies across languages, and how early children show ITL effects in perception. Iversen *et al.* (2008) show that in Japanese, manipulating intensity leads to an increased trochee rate just as in English, but the duration side of the ITL does not appear to apply. Bhatara *et al.* (2013) showed that for both intensity and duration, German speakers were more consistent in their ITL effects than French speakers. Such results suggest that the perception of sound sequences is influenced by language experience, and puts into question the status of the ITL as a psycho-physical 'law'. There are also results conflicting with a general parsing principle based on the ITL from production, in that intensity and duration can be reliable cues to stress both for trochaic and iambic stress in words (Revithiadou, 2004). Results from child studies furthermore suggest that while there is some evidence for an early intensity effect compatible with the ITL, the effect of duration appears to emerge later (e.g., Bion *et al.* 2011; Hay and Saffran 2012; Yoshida *et al.* 2010; see Crowhurst 2019 for a review).

Another issue with trying to leverage results of ITL studies based on tone and nonce-word sequences as a basis for a theory of speech segmentation is that the duration and intensity manipulations used in many ITL studies are often far greater than the acoustic differences observed in natural speech. For example, the intensity manipulation in Hay and Diehl (2007) went up to a 12 dB difference. Typical intensity differences due to stress in English tend to be much smaller, about half as large in the acoustic results reported in Wagner (2022) from productions of nonce-word sequences similar to those used Hay and Diehl (2007).[1]

In this paper, we present evidence for an alternative interpretation of the ITL proposed in Wagner (2022), one which, in contrast to the more recent interpretations Bolton's observation, does not involve direct reference to the notions of 'iamb' and 'trochee'. Rather, the hypothesis is that listeners parse the speech signal along two in-principle orthogonal dimensions: *grouping*, the segmentation of the signal into individual units such as words and phrases; and *prominence*, the identification of sounds that stand out as particularly salient. These correspond to basic dimensions of Gestalt perception. We spontaneously organize lower level sensory input into larger units (e.g., by the Gestalt principle of proximity), which is analogous to grouping the acoustic stream into words; and we spontaneously perceive some parts of the sensory input as figure and some parts as ground, which is analogous to the perception of prominence (Handel, 1989; Wertheimer, 1923).

The particular way in which intensity and duration cue grouping and prominence when it comes to parsing speech into words is that word-final syllables tend to be longer (Beckman and Edwards, 1990; Klatt, 1975; Oller, 1973; Wightman *et al.*, 1992)[2] and softer (Wagner, 2022) than word-initial syllables; and stressed syllables tend to be longer and louder than unstressed syllables (Beckman, 1986; Chrabaszcz *et al.*, 2014; Fry, 1958; Lehiste, 1970). In other words, the two cues correlate when they are used to convey word stress, and they anti-correlate when they are used to convey word segmentation (see Wagner and McAuliffe, 2019, for relating findings at the phrasal level).

If listeners use these cues accordingly in order to parse the speech stream, then the ITL, as stated by Bolton and assumed by many studies since, emerges as a generalization about edge cases in which either duration or intensity is manipulated in an extreme way: If I perceive a syllable as longer than expected based on its segmental content, I could attribute the excessive length either to finality or to prominence. If the excessive lengthening is so extreme that it cannot plausibly be attributed to one of these sources, I might come to the conclusion that the syllable has to be *both* stressed and final, and hence perceive an iambic word. Similarly, if I hear every other syllable as louder than expected given its segmental makeup, then I might attribute the excessive loudness either to initiality or to prominence. And if the excess loudness is greater than explicable by either of these inferences alone, I might hear them as *both* initial and prominent, and hence perceive the sequence as a series of trochees. These edge cases of extreme manipulations of one cue are informative, but focusing on them alone obscures the fact that they are just the tip of the iceberg of a much more general pattern. Under this view, the notions of 'iamb' and 'trochee' play no direct role in the explanation of the ITL, and we should not expect to find reliable cues for foot type as such.

In fact, we will see evidence that looking at the effect in terms of trochees and iambs is counterproductive, because there are in fact no consistent acoustic cues to foot type, other than in these edge cases. By asking listeners in some way or other whether they perceived the sound sequence as a sequence of iambs or trochees (e.g., Bhatara *et al.*, 2013; Boll-Avetisyan *et al.*, 2016, 2020; Hay and Diehl, 2007; Iversen *et al.*, 2008; Molnar *et al.*, 2016), earlier studies limited their own chances at finding consistent results. In these experiments, participants heard a sound sequence and had to say whether they heard them as repetitions of binary groups with initial prominence or binary groups with final prominence. For example, Hay and Diehl (2007, 115) asked participants "to indicate, by pressing a labeled button, whether the rhythm consisted of a strong sound followed by a weak sound or, alternatively, a weak sound followed by a strong sound." Bhatara *et al.* (2013, 3832) used a forced choice between depictions of trochees with a high bar followed by a low bar, and iambs with a low bar followed by a high bar, and a similar task was used in Iversen *et al.* (2008).

This is problematic, because these questions fail to establish the true percept, and confound prominence and grouping. Consider, for example, a sequence consisting of iterations of the syllables *out* and *look*. The sequence could be perceived as iterations of one of the four words *OUTlook*, *outLOOK*, *LOOKout* or *look OUT*, each with a different meaning, depending on which syllable is heard as initial within the word and which syllable within the word is heard as stressed (where capitalization marks the stressed syllable).

As illustrated in Table I, asking whether participants heard trochees or iambs neither settles the grouping question (is the syllable 'out' initial in the word, or the syllable 'look'?) nor the prominence question (is the syllable 'out' more prominent, or the syllable 'look'?):

	out initial	*look* initial
out prominent	(OUTlook) (OUTlook) (OUTlook) ...	OUT) (look OUT) (look OUT) (x...
look prominent	(outLOOK) (outLOOK) (outLOOK) ...	out) (LOOKout) (LOOKout) (X...

TABLE I. Four ways to parse a sound sequence alternating *out* and *look*.

Both *OUTlook* and *LOOKout* are trochees, and both *outLOOK* and *look OUT* are iambs. The notion of foot conflates these two dimensions.

Note that we use the term 'trochee' here simply to refer to a two-syllable sequence with initial stress and 'iamb' for a two-syllable sequence with final stress. Some phonological theories use a different, more technical notion of the terms 'trochee' and 'iamb', according to which the syllable *out* by itself would constitute a 'moraic trochee' in English (Hayes, 1995). We will return to this in the final discussion.

Many earlier studies of the ITL either used alternating tones of equal frequency (e.g., Bell, 1977; Rice, 1992; Vos, 1977) or used sequences of identical syllables such as *gaga* (Hay and Diehl, 2007), where this issue with the 'foot task' may have been less obvious. Another reason why the problem may not have been considered important is that in most earlier studies, only one cue was manipulated at a time, so every other syllable may have been louder or longer, but not both; the experiments also didn't include stimuli where one syllable was louder and the other longer. If listeners always hear the longer/louder syllable as prominent, then in a way the foot task would give us complete information: If a listener heard a trochee, and we can be sure that the longer/louder syllable is heard as stressed, then we can actually identify the cell in the table in Table I. As we will see, however, there is a lot of variability in whether the manipulated syllable is actually heard as stressed, because intensity and duration can also be attributed to initiality/finality instead of being attributed to prominence.

Crowhurst (2016) also discusses this potential confound, and used a speech segmentation task instead of the foot decision task, which establishes the grouping perception, following earlier work on the ITL from the acquisition literature. Yoshida *et al.* (2010) and others in fact used related tasks in their study of the acquisition of the ITL, because it is much easier to establish grouping perception in children than foot perception, since one cannot ask an explicit question to establish foot type (see also Abboub *et al.*, 2016; Bhatara *et al.*, 2016; Bion *et al.*, 2011; Hay and Saffran, 2012; Molnar *et al.*, 2014; Yoshida *et al.*, 2010, for studies of the ITL using speech segmentation tasks). A speech segmentation task, however, does not answer the question of where prominence was perceived, and thus does not fully establish the precise percept either. In order to understand the ITL, we have to look at the perception of both grouping and prominence.

Initial evidence in favor of the view that the ITL is an indirect consequence of the parsing of the speech signal along these two dimensions was presented in Wagner (2022), based on production and perception of tone sequences and speech sequences in English. The production evidence showed that initial and stressed syllables in English bisyllabic nonce words are louder, and that final and stressed syllables are longer. There was no evidence that duration is specifically increased to cue iambs, or intensity to cue trochees. Rather, apparent differences between foot types are mostly just additive effects of grouping and prominence: In iambic words, stressed syllables look particularly long because the effects of final lengthening and stress lengthening add up; in trochaic words, stressed syllables look particularly loud because initial syllables within a word are louder than final syllables and stressed syllables are louder than unstressed ones.

The perception experiments in Wagner (2022) show that listeners indeed treat an increase in intensity as a cue to initiality or stress, and duration as a cue to finality or stress. Increasing the loudness of a syllable is hence ambiguous: It could convey stress or initiality. And an increase in duration is ambiguous between cueing stress or finality. Duration and intensity were very poor cues to foot type in this study. Further evidence for this two-dimensional parsing hypothesis is reported in Wagner *et al.* (2021), based on a cross-linguistic study of speech perception in 6 languages.

The present paper replicates some of the earlier results from Wagner (2022) and Wagner *et al.* (2021), specifically that listeners interpret duration as a cue to prominence and finality within a group, and intensity as a cue to prominence and initiality, and also confirms that intensity and duration are indeed very poor cues to foot type. The ITL as originally stated is only observed under specific circumstances, when either duration or intensity is manipulated in an extreme way, where neither grouping nor prominence alone can explain the degree of the difference. Studies using the foot decision task hence restrict themselves to studying the exceptional instead of the general.

The present study also goes beyond earlier results in several ways. First, all prior tests of the ITL have looked at sequences of nonce words, while the present experiments look at sequences of actual words. This is an important step to ensure that the generalizations found might plausibly generalize to more naturalistic speech.

Second, we illustrate not just that there are no consistent cues to the distinction between iambs and trochees but also *why*: We show that two acoustic cues actually cue the dimensions of grouping and prominence, and that these cues to the distinction between trochees and iambs are consistent only after we condition on one of these other dimensions.

The third way in which the present results go beyond prior studies is that we explore how exactly decisions about prominence and grouping inform each other. A recurrent finding in the literature on the ITL is that of a trochee bias, which suggests a relationship between prominence and grouping perception. Jakobson *et al.* (1951) speculated that such biases in the perception of tone sequences are a function of the dominant lexical stress pattern in a language. Consistent with this idea, Cutler and Butterfield (1992); Cutler and Carter (1987); Cutler and Norris (1988) and Mattys *et al.* (1999) show evidence that stress is taken by English listeners as a cue that the syllable initiates a new word. Following Cutler and Carter (1987), such effects are often attributed to the lexical statistics of English, pointing to the observation that 70% of bisyllabic words in English have initial stress. Compatible with the claim of a lexical source, Bhatara *et al.* (2013) found a trochaic bias in German but not in French in their control stimuli. However, the evidence was somewhat equivocal: While German showed a higher trochee bias than French, French listeners *also* showed a weak trochee bias, an effect not expected given the prosodic pattern of French words and phrases, which have final prominence. Similarly, Wagner *et al.* (2021) found a trochee bias in their control stimuli in English, German, and French alike. It is hence possible that there is also a language-independent bias toward perceiving initial prominence, possibly the same that is responsible for the perception of an initial downbeat in music, which decides prominence when other factors fail to do so (Povel and Okkerman, 1981).

Another potential source of mutual influence of the two decisions is that since the cues used to encode the two dimensions overlap, perceiving a syllable as prominent might modulate the acoustic effects on the grouping decision and vice versa. If part of the intensity and duration is 'explained away' as being due to stress, this should have the effect that this portion of the acoustic information will no longer need to be explained by grouping. Such perceptual modulation effects are indeed reported in Wagner (2022), but this interaction between the 'other' decision and the effect of the acoustic cues was quite small.

In order to further explore these questions of how the two decisions mutually inform each other, we manipulated the precise task that participants engaged in. One group of participants answered a grouping question first ("did you perceive XY or YX?"), and then a prominence question ("is X or Y stressed?"), on each trial. A second group of participants answered the prominence question first, and then the grouping question. A third group of participants was given four randomly ordered choices (i.e., answered the prominence and grouping questions together).

The idea behind this task manipulation was that the task itself might modulate the mutual influence of the decisions on each other by changing which perceptual decision is made first. It could be, for example, that whichever decision is made first is made independently, and then the second decision accounts for the cues not already 'explained away' by the first decision. This would constitute an 'early-bird' effect, where the first perceptual decision is credited for any ambiguous information in the signal. However, it could also be that the task is irrelevant, and cognitively it is always one of the two decisions that is made first. The results show clear evidence that the task did indeed change how the two decisions affect each other, but not in a way we anticipated: The decisions affect each other more when one asks for the prominence-percept first. We explore possible reasons in the final discussion.

The final way in which this study goes beyond prior studies is that it uses a more adequate statistical model to test these effects. Wagner (2022) used two separate binary logistic mixed effect regression models to analyze the two decisions and the mutual effect on each other. But as noted in that paper already, this analysis treats the two decisions as independent, while they are clearly not, because they were taken on the same trial by the same participant, and the very results of the models showed that they did influence each other. The models also made mutually incompatible assumptions about the direction of causation: Letting two decisions influence the other creates a chicken-and-egg problem in the causal assumptions underlying the model. These issues are addressed here by using multinomial regression models, which have the advantage that the model does not force us to prejudge the direction of the mutual effects in the specification of the model itself. Rather, we can study the mutual effects, and possible causal structures, by exploring the model's predictions.

II. METHODS

We conducted three online experiments, varying in task. The experiments were run on a McGill server, using Javascript. We implemented the experiments with the prosodylab-experimenter,[3] a set of javascripts which in turn makes use of jsPsych (De Leeuw, 2015). The experiment was preregistered on OSF.[4]

Each experiment had four blocks, one for each of our four item sets. Each item set involved sequences of two syllables, which can combine to create 4 different words of English, each with a different meaning (*out* and *look*, *set* and *in*, *set* and *up*, *turn* and *over*). For instance, the syllable pair *set* and *up* can be combined to form *SETup*, *set UP*, *UPset*, and *upSET*, depending on whether *set* is initial or *up* is initial, and whether *set* is stressed or *up* is stressed. Some of the words were infrequent, or even productively formed. For example, *outLOOK* is not listed as a word in Webster's dictionary, although it is in the OED (last mention 1994). But *out*-prefixation is a productive morphological process, and even without prior exposure the word could be used in statements such *This car out-looks and out-performs its competition*. The prefix *out-* serves as a comparative here, conveying that someone looks better than someone else, or maybe looks faster at something than someone else. In order to reduce effects of word familiarity, we explained the meaning of the words in the instructions for each of the four blocks of the experiment. The syllables were generated using Amazon Polly Text-to-Speech synthesizer.[5] All syllables were generated using the 'Joe' voice with the language set to English US, and the sampling rate set to 22050 Hz. Using Praat (Boersma and Weenink, 2020), each pair was acoustically normalized in intensity, duration, and pitch. The pitch for each syllable in the pairs was uniformly set at 100Hz across all item sets. The baseline levels of intensity and duration, however, were not set uniformly, but adjusted based on our perception such that the baseline stimuli were maximally ambiguous. A uniform setting of duration and intensity for all syllables would not work since some segments are

Item	Syllable	Intensity (db)	Duration (ms)	Item	Syllable	Intensity (db)	Duration (ms)
out-look	out	70	0.320	set-up	set	70	0.340
	look	70	0.360		up	70	0.220
set-in	set	70	0.360	turn-over	turn	70	0.320
	in	70	0.220		over	70	0.320

TABLE II. Baseline levels of intensity and duration for each syllable in the item sets.

inherently longer and louder than others, and listeners will correct for their expected length and loudness in perception. The idea was that in the baseline sequences, the speech stream should be four-way ambiguous (e.g., between *SETup*, *set UP*, *UPset*, and *upSET*). After trial and error, we ended up with the values given in Table II. The syllables in each pair were then concatenated, with a 5ms pause in between, forming two words, one for each possible order. For instance, *out* and *look* were concatenated in both possible orders to form two words, *outlook* and *lookout*.

Each word was then manipulated, using a Praat script. Manipulation only targeted the steady state of the vowel of the syllables in each word. There were two levels of intensity/duration variations, 0 and 9dB above the baseline for intensity and 0 and 120ms above the baseline for duration, which are relatively high compared to differences observed in natural speech, but small enough to avoid streaming effects and unnaturalness. We followed Crowhurst (2016) in that each cue was only manipulated on one syllable at a time, but both cues might have been manipulated on the same or on different syllables of a given word.

For instance, if one syllable was manipulated in intensity, the other syllable was always set at the baseline level for intensity, but either syllable could be manipulated in duration. This resulted in a total of 9 different manipulations (1 baseline with no manipulation, 2 intensity-only manipulations in which one or the other syllable was louder, 2 duration-only manipulations, and 4 stimuli where both duration and intensity were manipulated, either on the same syllable or different syllables).

For each item set, this was done twice, once for each underlying syllable order. For example, there were 9 *out-look* and 9 *look-out* stimuli. The underlying syllable order has been found to affect foot decisions in ITL experiments (Hay and Diehl, 2007), so it was counterbalanced within each participant. Results in Wagner (2022) and Wagner *et al.* (2021) suggest that underlying order specifically affects the grouping decision, but not the prominence decision. This makes sense, since listeners will tend to assume that the first syllable emerging from the noise is the beginning of a new word.

So for each item set, there were 18 bisyllabic words, 9 for each ordering of the two syllables that varied in the acoustic manipulation. Each manipulated bisyllabic word was then concatenated 12 times to form a sequence, resulting in 18 different stimuli for each item set.

To minimize the effect of underlying order, we added a 0.5 sec on-ramp at the beginning of the sound file, based on a quarter-sine ramp with superimposed white noise fading out over the same time frame. We added a parallel off-ramp with a white noise fading in at the end, to avoid that listeners assume that the last two syllables form a word. Stimuli were converted to mp3 format in order to reduce file size. Every participant listened to 4 blocks of 18 stimuli, one block for each item set. The four blocks were ordered randomly between participants. We counterbalanced block order such that between participants, each block was about equally likely to occur first.

At the beginning of each block, participants were instructed to listen to sequences consisting of repetitions of one of the four words/phrases each block corresponds to. They were also given an example and definition for each of the four English words.[6]

There were three groups of participants, corresponding to the three different tasks. One group answered a two-way forced-choice grouping question first, and then answered a two-way forced-choice question about which syllable they thought was stressed. For instance, they might have listened to a sequence like *...outlookoutlook...* the first question gave them two choices, *look-out* or *out-look*, and they were asked: "In the word/phrase that you heard, did 'out' come first (as in out-look), or 'look' (as in look-out)?"

If they answered *out-look*, the second question gave them the choices *OUTlook* and *out-LOOK*, and asked: "Where did you hear stress? (stress is marked by capitals)" We randomly varied which choice for each of the questions was presented first (e.g., *look-out* or *out-look* for the grouping choice) between participants, to prevent any bias due to overall order of the presentation of the choices in the questions.

A second group of participants answered the prominence question first, for example, they were asked whether they heard *out* or *look* as stressed. If they responded *out*, they were then asked whether the word/phrase they heard repeated was *OUTlook* or *look OUT*. Otherwise the experiment was identical to the Grouping-first version.

These two tasks differing in question order were the ones that we originally planned at the time of preregistration. After running these two experiments, we added a third task for a third group of participants, in which participants chose one of the four choices in a four-way forced-choice task. For this version of the experiment, the four choices were randomly ordered anew on each trial, to prevent that the order of the four choices lead to a Grouping-first or Prominence-first bias across the experiment. The goal of this third task manipulation was to establish the response pattern when participants were not told along which dimensions to parse the signal, and no 'order' between making any decisions was preconfigured by the task.

A total of 155 participants were recruited for an online study on Prolific, 50 for each of the two question orders, and 55 for the Four-way choice task. We restricted participation based on Prolific filters to monolingual English speakers, and to have been born and currently reside in Canada or the US, as well has having an approval rating of 98% or higher on Prolific tasks. Participants completed a language background questionnaire, and did a sound check for their headphones and mic before the experiment. Despite the Prolific filter we used, we had to exclude 2 participants because they reported not being native speakers of North American English.

Given past online experiments in our lab, we expected a small proportion of participants would report being monolingual English speakers on Prolific and our language questionnaire, even though they are not native English speakers. To remove non-native speakers from the pool, we had each participant conduct a mic check for a production task that followed the perception experiment, as specified in the amendment to the preregistration (which we added before data collection began). For the mic check, we recorded each participant saying the sentence *The cat followed the squirrel stealthily to its nest.* Based on their production of this sentence, which contains various challenges for non-native speakers, we excluded 12 participants, either because their English sounded non-native, or because the recording did not work and we couldn't check their native speaker status.[7]

Participants were instructed to use headphones for the study, and do the study in a single sitting, without interruptions and distractions. In order to check whether the participants were actually wearing headphones as instructed, we used a slightly adapted version of the headphone screening test from (Woods *et al.*, 2017), which they could only pass if they were (i) wearing headphones, and (ii) paying attention. After excluding 28 participants who failed the headphone screening test, a total of 113 participants remained (34 for the Grouping-first task, 33 for the Prominence-first task, and 46 for the Four-way choice task). This is a higher number than the number of participants tested in Wagner (2022), a study which nevertheless proved powerful enough to detect relevant effects. More information about the participants and the headphone screener task is given in Supplemental Information.

After the experiment, participants answered a short questionnaire in which they reported what they thought the experiment was about, and they were also given the opportunity to report problems they encountered while running the study, if any.

III. RESULTS

A. Exploratory Data Analysis

The heat maps in Fig. 1 show the empirical data for the prominence decision (first row) and grouping decision (second row) for each of the three task manipulations (Prominence-first, Grouping-first, and Four-Way-Choice). For each item set, we arbitrarily call one of the two morphemes X and the other Y, for example, for the *look-out* item set *look* always corresponds to X and *out* to Y. The color of each cell within a heat map encodes the proportion of responses for X being prominent ('X') or X being heard as initial within a word ('XY'), depending on duration and intensity.

Each cell averages over an equal number of observations from each 'underlying' order, e.g., an equal number of cases in which *out* was underlyingly first in the sequence and in which *look* was underlyingly first. Balancing underlying order is crucial to control for left-to-right biases in perception, especially of grouping.

Note that for the Four-way forced-choice task, participants chose between four options, but here we nevertheless plot the resulting responses as a function of prominence and grouping perception (where 'Prominence' = the proportion of responses with choice = X/XY or X/YX, and so on), in order to allow for comparisons across tasks.

a. Acoustics effect on prominence and grouping decision. The prominence plots (the three heat maps in the top row) show that as predicted, increasing duration and intensity each made it more likely that a syllable was perceived as prominent, with the most consistent responses in the top right and bottom left corners in each of the three heat maps. If one

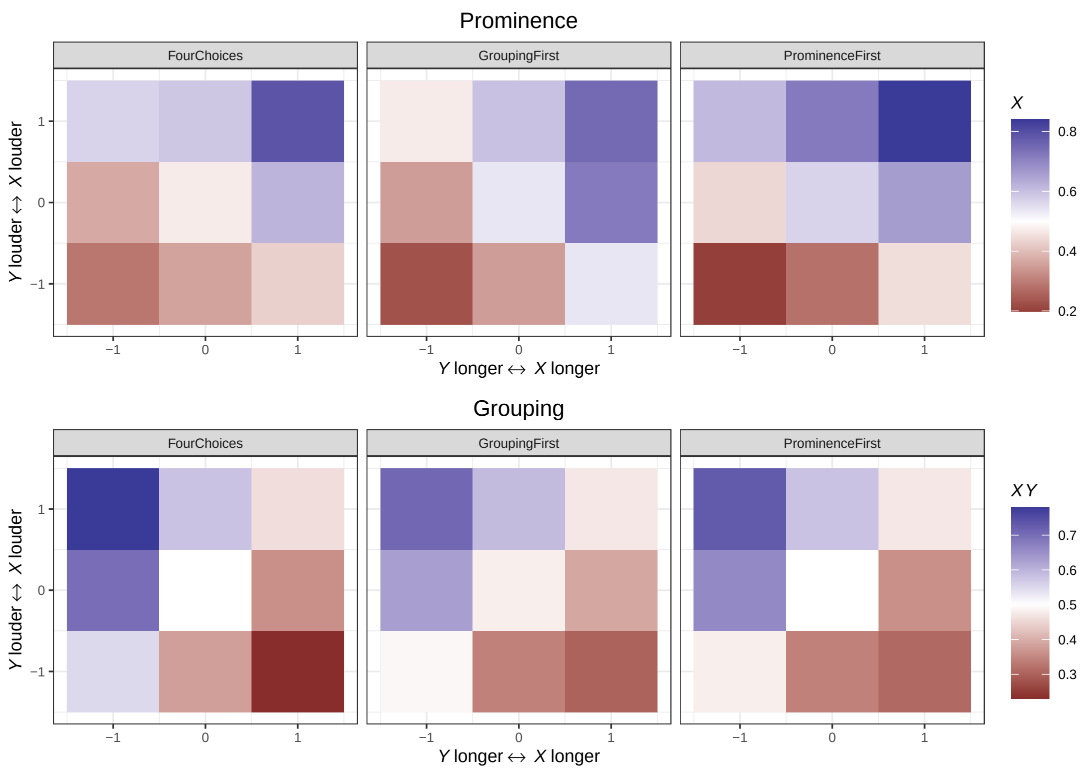

FIG. 1. Proportion of responses for each of the three task manipulations depending on intensity and duration manipulation for the prominence decision and the grouping decision. The bottom left corner of each heat map plots the responses for stimuli where X was shorter and softer than Y, the top right corner plots the responses for stimuli where X was longer and louder than Y, and so on, with the responses to the baseline stimuli in the center.

syllable was louder and the other longer, responses were closer to chance, leading to a 'diagonal of uncertainty' from the top left to the bottom right of the prominence heat maps. There were some subtle differences between tasks in the cue weighting, for example in the Prominence-first task, intensity seems to have a played a greater role in the prominence decision than in the Grouping-first task.

The row of plots for the grouping decision shows that as hypothesized, softer and longer syllables are more likely to be perceived as final, and louder and shorter syllables as initial. Consequently, responses were most consistent in the top-left and bottom-right corner of the grouping heat maps. When the acoustic cues gave inconsistent cues for grouping, i.e., when the same syllable was both longer (and hence cued to be final) and louder (and hence cued to be initial), as in the top-right and bottom-left corners of the grouping heat maps, responses were closer to chance. This leads to a 'diagonal of uncertainty' orthogonal to the one in the prominence plots, from the bottom left to the top right of the heat maps. Again, there were some subtle differences between tasks in the cue weighting, such that in the Four-choices task, intensity seems to have played a slightly greater role in the grouping decision than in the the other two tasks, which may however just be due to random variation (we have no other explanation).

b. Comparison of the four item sets. Fig. 2 illustrates that the pattern of how intensity and duration affected the prominence and grouping decisions was quite consistent across the four different word quadruples. When looking at the individual results for each item set, as illustrated in Fig. 2, we see the same qualitative pattern across all item sets, which vary only in their overall bias toward hearing one of the two morpheme as initial or hearing one of the two morphemes as prominent.

c. Effect of task order. The order in which the questions are asked might influence the direction of the mutual effect of one decision on the other. One possibility is that we will see an 'early-bird' effect, such that whichever dimension we ask about first will be made independently and get the credit for whatever aspects of the cues it can account for. If

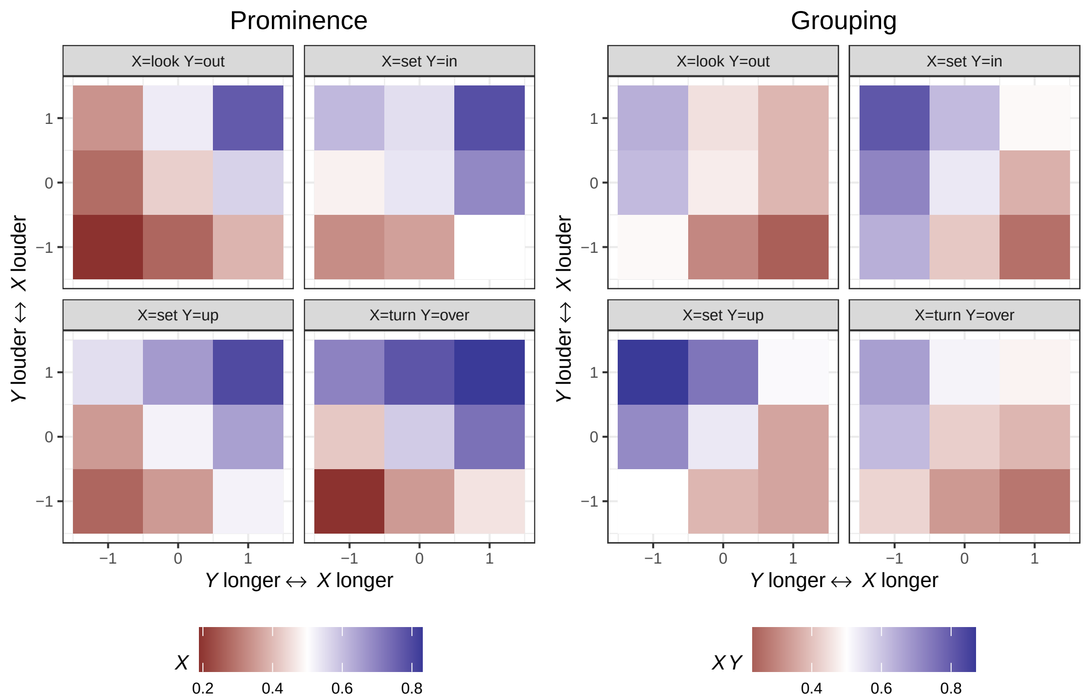

FIG. 2. Proportion of responses for the grouping and prominence decisions, depending on intensity and duration manipulations, for each of the four item sets.

true, then in the Prominence-first order, the prominence decision should be made relatively independently, and it should tend to influence the grouping decision. And in the Grouping-first order, the grouping decision should be made relatively independently, and it should tend to influence the prominence decision. We can explore whether this is the case by faceting each decision by the other decision, and evaluating how different the heat maps are depending on the task order.

Fig. 3 shows the plots for the prominence decision faceted by task order and the grouping decision. The 'early-bird' hypothesis would predict that acoustic cues are more likely to be attributed to prominence (rather than grouping) in the Prominence-first order. In some

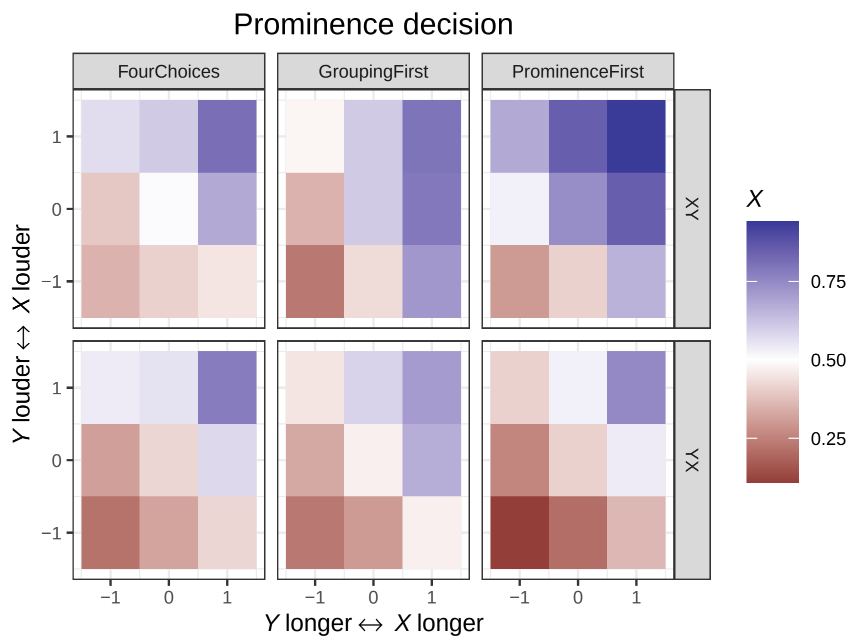

FIG. 3. Proportion of responses for the prominence decision, conditioning on the grouping decision, as a function of intensity and duration manipulations.

sense the plots are compatible with this, since two heat maps in the final column show the greatest effects, judging by the shading of the cells in the extreme bottom-left and top-right corners. The specific direction of the effect, is that if X was perceived to be initial (‘XY’), then X was more likely be perceived as stressed (the top heat map is shifted more toward blue compared to the bottom heat map). In other words, the Prominence-first order seems to have the effect that it makes listeners more likely to report hearing trochees (rather than just showing greater acoustic effects on the prominence decision).

Fig. 4 shows the grouping responses faceted by the prominence decision. The early-bird hypothesis would predict that we should see greater acoustic effects in the Grouping-first order, since grouping should be more likely to get the credit, but the opposite is the case.

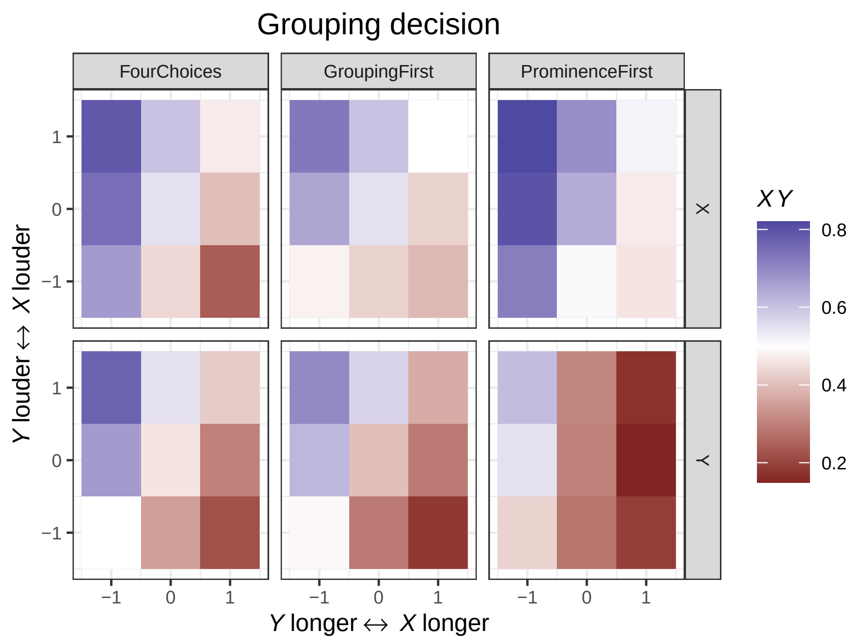

FIG. 4. Proportion of responses for the grouping decision, conditioning on the prominence decision, as a function of intensity and duration manipulations.

Here too, we see that the heat maps in the Prominence-first order show the greatest differences. And here too, the differences reveal a greater trochee rate: When X was perceived to be prominent, it was more likely to be perceived as initial in the Prominence-first order, as is visible in the shift to the blue color palette in the top heat map for the grouping decision, compared to the heat map below, which is shifted more toward red, which means that Y was more likely to be perceived as initial if it was heard as prominent.

The effect of task was then consistent across both decisions: For both grouping and prominence decisions, the two decisions had a greater influence on each other in the Prominence-first order, resulting in a greater trochee rate. The Grouping-first and the Four-way choice tasks showed a smaller trochee bias and looked more similar to each other. For both deci-

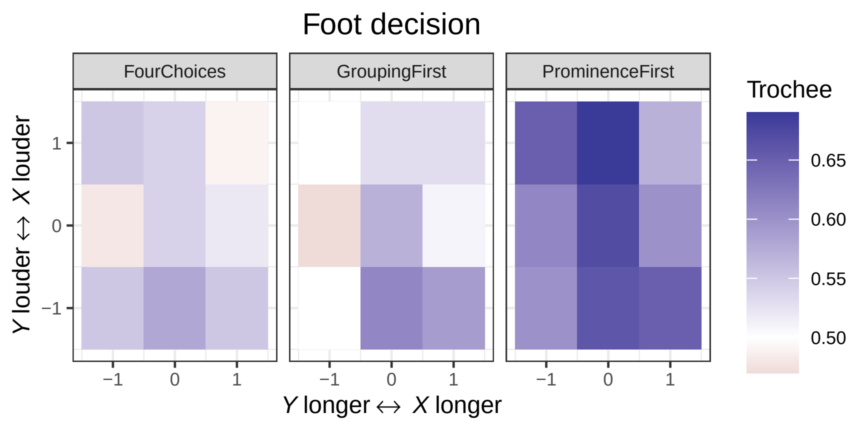

FIG. 5. Proportion of trochee responses for the (reconstructed) foot decision for each of the three task manipulations, depending on intensity and duration manipulations.

sions, the mutual influence biases toward trochees: If X was perceived as prominent it was then more likely to be perceived as initial, and if X was perceived as initial it was then more likely to be perceived as prominent. This modulation of the trochee rate is easier to see when looking at the plots of which foot type the participants heard in the signal.

d. Acoustic and task order effects on the foot decision. While we did not directly ask participants about which foot type they heard, we can reconstruct this information from the grouping and prominence decisions, and also from the choice they make in the Four-way task. For example, if a listener heard *set* as initial and prominent in the *set-up* item set, then their foot decision was coded as a trochee. Fig. 5 plots the reconstructed foot decisions for each task type.

One salient difference between the plots of the foot decision and the other decisions seen above is that the foot heat maps are comparatively mono-chromatic. While for grouping and prominence intensity and duration shifted the proportions in perception through most of the probability space, neither cue had a great influence on whether participants heard iambs or trochees.

The manipulation of the task, on the other hand, had a very clear effect on the foot decision: While there was only a slight overall trochee bias in the Four-way and Grouping-first task, the heat map for the Prominence-first task shows a much stronger trochee bias, as reflected in an overall bluer coloring of the heat map. This confirms what we already saw when looking at the grouping and prominence decisions conditioned on the other decision: Asking for prominence first leads to a greater trochee rate.

Looking more closely at the foot decision, we can see that there was also a trend for the ITL pattern: When the syllables only differ in duration but are equal in intensity, as in the marginal cells of the middle rows of the heat maps, the rate of iambic responses is slightly higher compared to the baseline in the central cell in all three foot decision plots. This provides some evidence for the duration side of the ITL.

There is less evidence for the intensity side of the ITL: When the syllables only differ in intensity, as in the marginal cells of the middle columns of the three heat maps, we see that the trochee rate is higher compared to the baseline stimulus only in half of the cases.

Overall, the effects of the acoustic manipulation on the foot decision were clearly very small. Why is this the case? Note that among the three decisions (grouping, prominence, foot type), any subset of two decisions encodes the exact same information, and disambiguates the four-way contrast. So should the foot decision not be similarly informative to the other decisions, and should there not be consistent cues for foot type as well?

To understand better why intensity and duration are such poor cues to foot type, it is important to take into account that the notions 'iamb' or 'trochee' inherently build on the notions of prominence and grouping. We can see how this results in poor differentiation by acoustic cues by faceting the foot decision either by the Grouping decision, or by the Prominence decision, as we have done in Fig. 6 (averaging across all three tasks).

Let's first look at the foot decision faceted by the prominence choice. It is clear that the direction of the acoustic effects are reversed depending on whether listeners heard X as prominent or Y. This makes perfect sense given our hypothesis for the cue interpretation for the dimension of grouping: If the listener is already committed to X being the prominent syllable, then any increase in duration of X will be taken as a cue that X is final, leading to the conclusion that stress is iambic, and any increase in intensity of X will be taken as a cue that it is initial, hence the word will likely be heard as trochaic. If on the other hand, the listener has already decided that Y is prominent, then the inferences about grouping will be exactly parallel, but their effect will be the opposite: An increase in duration of X will again be interpreted as a cue that X is final, but this time will result in a trochaic percept. Any increase in the intensity of X will again be heard as a cue that it is initial, but this time the effect will be that the perceived word has iambic stress.

There is a similar mirror-image pattern when we condition foot choice on the two different grouping choices. When the perceived grouping is XY, any increase in duration or intensity of X will increase the likelihood that it is heard as stressed, and hence a trochaic pattern is perceived. When the perceived grouping is YX, then any increase in duration or intensity of X will again be attributed to stress, but in this case will lead to more iambic responses.

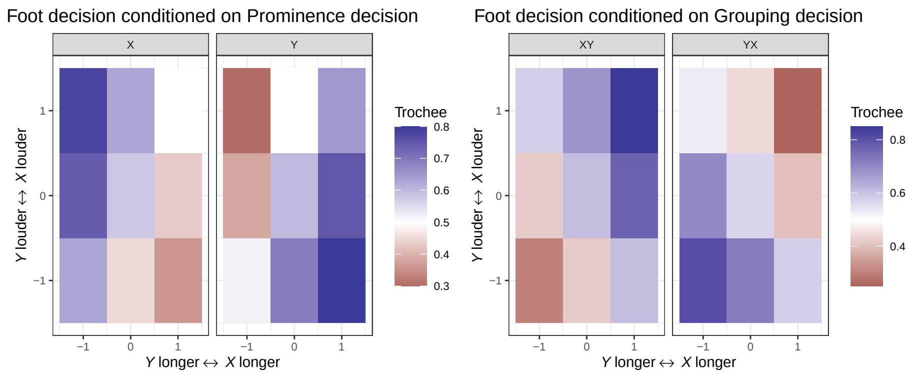

FIG. 6. Foot decision conditioned on prominence decision (left) and grouping decision (right). The effects of duration and intensity for each decision are reversed depending on the other decision.

These plots illustrate clearly why there are no reliable cues to foot type: The cues actually reflect grouping and prominence, and their effect will have consistent effects of perceived foot type only if we hold either their grouping or prominence choice constant.

B. Statistical model

To quantitatively model participants' behavior and statistically test the observations made from empirical data above, we model `PG`, the choice participants make (combination of `Prominence` and `Grouping`), using a multinomial logistic regression. There are four possible outcomes for `PG` in each trial, corresponding to each combination of the prominence (X, Y) and grouping (XY, YX) decisions. We write these as follows (using the item set 'out-look' for illustration):

1. X.XY (e.g., OUT-look)

2. Y.XY (out-LOOK)

3. X.YX (look-OUT)

4. Y.YX (LOOK-out)

In describing our model, we assume familiarity with frequentist mixed-effects (binary) logistic regression models, for example using the `glmer()` function from the `lme4` R package. Wagner (2022) analyzed very similar data using two mixed-effect logistic regressions, one for the prominence decision and one for the grouping decision. This treats the two decisions as independent choices by participants, which is arguably not justified in any of our three task types, most obviously in the Four-way choice.[8]

Our regression models `PG` as a function of the following population-level predictors ('fixed effects'): duration (`Longer`), intensity (`Louder`), underlying syllable order (`SyllableOrder`: levels XY, YX), task order (`TaskOrder`: levels FourChoices, GroupingFirst, Prominence), and item set (`item`: levels LOOK-OUT, TURN-OVER, SET-UP, SET-IN). To allow for the possibility that the effect of acoustic cues differ by task order (Fig. 1), the model contains interactions of `TaskOrder` with `Longer` and `Louder`. These terms account for all research questions, and all salient effects observed in the empirical data and in Wagner (2022)'s data. While additional interaction terms would be theoretically possible (e.g., a `Longer`:`Louder` interaction or `SyllableOrder`:`TaskOrder` interaction), we did not include these in the absence of clear theoretical motivation and to avoid overfitting.

The group-level structure of the model ('random effects') allowed for all logically possible by-participant variability, in the intercept as well as the `SyllableOrder`, `Louder`, and `Longer` effects, with all possible correlations between different kinds of variability. Although it is not actually possible to fit a multinomial model in `lme4`, we can use `lme4` notation to describe the fixed and random effects in a model formula:[9]

```
PG ~ Item + SyllableOrder + TaskOrder * Louder + TaskOrder * Longer + (1
+ SyllableOrder + Louder + Longer | participant)
```

Predictor `TaskOrder` was coded using Helmert contrasts, and `SyllableOrder` was coded as -0.5/0.5 (orders XY/YX). Predictors `Louder` and `Longer` each had 3 values: -1 (Y longer or louder; X at baseline), 0 (X and Y at baseline), and 1 (X longer or louder; Y at baseline). `Louder` and `Longer` were then coded as ordinal factors, using *monotonic effects* (Bürkner and Charpentier, 2020), which allows each predictor to affect participant decisions in a consistent direction. For example, our hypothesis predicts that a higher value for `Louder` should make outcomes more likely in which X is prominent (X.YX AND X.XY) and outcomes in which X is initial, including Y.XY; it should make the outcome Y.XY less likely, since it cannot explain higher intensity on X. By coding `Louder` as an ordinal predictor, we do not assume that each step along the continuum ($-1 \rightarrow 0$, $0 \rightarrow 1$) will the same effect. For our purposes, this means that there are `Louder` and `Longer` coefficients which can be interpreted intuitively ("the effect of `Louder`"). (Also fitted are two thresholds, in this case $-1/0$ and $0/1$, which we do not discuss.) Details of the model are given in Supplemental Information Sec. III.

To actually fit the model described above, we opt for a Bayesian regression model. (See McElreath, 2020; Nicenboim *et al.*, 2022 for introductions; and Vasishth *et al.*, 2018 for phonetic data in particular.) We use a Bayesian model for practical rather than philosophical reasons: Bayesian models allow for greater flexibility in model specification, and are easier to fit and interpret for complex model structures, compared to frequentist models. While it is possible to fit frequentist mixed-effects multinomial logistic regressions (e.g., Davidson, 2016; Sonderegger *et al.*, 2020 for phonetic data), existing packages only allow for restricted model structure[10] and are harder to fit and interpret. The key difference between Bayesian and frequentist models, for our purposes, is that Bayesian models result in probability distributions of likely values of model parameters given the data ('posterior distributions'), where frequentist models result in estimates of model parameters, whose uncertainty can be quantified with confidence intervals or p-values. We only summarize posterior distributions using quantities analogous to frequentist models (credible intervals, p_d values), so our analysis should be legible even with minimal knowledge of Bayesian models.

We fit a Bayesian version of the mixed-effects multinomial logistic regression described above using `brms` (Bürkner, 2018: version 2.16.3), an R-based front-end to the Stan programming language (Stan Development Team, 2021; Team *et al.*, 2019) (`categorical` family, with logit link). Analogously to how a binary logistic regression (with two levels) fits the relative probability (as log odds) of one outcome (1) versus another (0), a multinomial logistic regression for k outcomes fits $k-1$ probabilities, of outcome $i+1$ versus outcome 1 (the 'base level').

Because `PG` has four levels ($k = 4$), our model fits three probabilities (as log-odds): P(X.XY vs. Y.XY), P(X.XY vs. X.YX), and P(X.XY vs. Y.YX), as a function of the predictors and group-level structure above. These probabilities are not intuitively interpretable in terms of our research questions, which is why we will unpack the model results by calculating more interpretable quantities. We return to model interpretation below.

Getting the 'results' of a Bayesian model means drawing many samples from the posterior distribution once the fitting algorithm has converged, repeating the process across several 'chains' to ensure convergence (all chains should give similar results). Our model was fit using 18000 samples across 6 Markov chains (1500/1500 warmup/sample split per chain), which were sufficient for convergence (see Supplemental Information for model diagnostics). Fitting the model involves specifying prior distributions for all parameters, which can be used to capture information from previous studies, or to aid model fitting. We used weakly-informative regularizing priors (McElreath, 2020; Nicenboim *et al.*, 2022), which do only the latter: they aid model convergence by discouraging unrealistically large effects, while providing no information about effect direction. Prior choice is discussed further in Supplemental Information Sec. III.A. The fixed effects used Normal(0,3) priors for regression coefficients and Dirichlet(1) priors for threshold parameters of the monotonic effects. Group-level parameters used the `brms` default of a half Student's t-distribution with 3 degrees of freedom and a scale parameter of 10 for variances; correlations used an LKJ prior with $\eta = 1.5$, which gives lower probability to perfect (-1/1) correlations. These are all very weak priors, by the metric of effects in (Wagner, 2022).[11]

a. Interpreting the model. Multinomial regression models are difficult to interpret from the estimated model coefficients, as is usually done for mixed-effects models (e.g., when interpreting `lme4` models), because the coefficients of multinomial models do not correspond straightforwardly to quantities of interest. It is easier to interpret the model by examining its predictions for quantities of interest specifically calculated from the model predictions to address particular research questions (e.g., McElreath, 2020, 11.3). This is analogous to examining custom model predictions from frequentist models using 'estimated marginal means' or post-hoc tests (e.g., Sonderegger, 2023, Chapter 7), when this is easier than interpreting model coefficients and p-values.

In our case, we are not directly interested in the three probabilities estimated by the model (X.XY vs. Y.XY, etc.), but instead probabilities which we denote $P(X)$, $P(XY)$, and $P(\text{trochee})$:

- $P(X)$: the probability that `Prominence` is X (decision 1)
- $P(XY)$: the probability that `Grouping` is XY (decision 2)
- $P(\text{trochee})$: the probability that the resulting foot is a trochee.

Each of these corresponds to two possible outcomes for `PG`:

$$P(X) = P(\texttt{prominence} = \text{X}) = P(\text{X.XY}) + P(\text{X.YX})$$

$$P(XY) = P(\texttt{grouping} = \text{XY}) = P(\text{X.XY}) + P(\text{Y.XY})$$

$$P(\text{trochee}) = P(\texttt{foot} = \text{TROCHEE}) = P(\text{Y.YX}) + P(\text{X.XY})$$

Because the model predicts the probability of each outcome of `PG`, given values of the predictors, we can use the model to get predictions of any quantity of interest z involving these probabilities: in our case, we will usually interpret quantities involving $P(X)$, $P(XY)$, or $P(\text{trochee})$. For example, z might be "how much $P(X)$ changes as `Louder` increases" (the "effect of `Louder` on the prominence decision").[12]

To interpret a Bayesian model, and what it tells us about our research questions, we examine posterior distributions, which describe how likely different values are (of a model

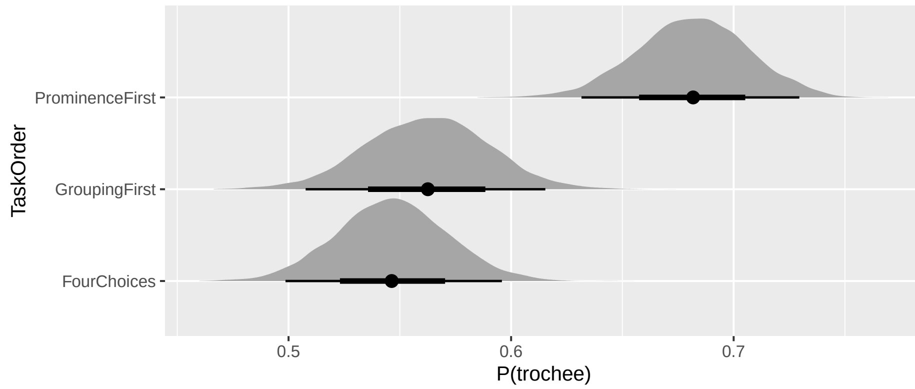

FIG. 7. Model-predicted posterior distribution of P(trochee) by task order, with median (dots) and 66/95% credible intervals (lines).

parameter, or a quantity of interest computed from model parameters) given the observed data. For this model, we use posterior distributions for each quantity of interest z. For example, Fig. 7 show the posterior distribution for P(trochee) for each level of `TaskOrder`: this is the predicted probability that the outcome is X.XY or Y.YX ('trochee'), for an average participant and averaging over `SyllableOrder`. It is visually clear that P(trochee) is larger when `TaskOrder` = ProminenceFirst than other task orders, reflecting the empirical distribution in Fig. 5. Thus, one z discussed below will be "difference between P(trochee) for ProminenceFirst vs. other task orders".

To interpret the posterior distribution for each z, we must somehow summarize what it says about which values of z are likely and what this means for our research questions. We use three simple and common metrics to summarize posteriors: the median and 95% credible interval (CredI: the interval between the 2.5% and 97.5% quantiles), and probability of the effect's direction, written p_d (Makowski *et al.*, 2019). These quantities are roughly analogous to summarizing a predicted effect for a frequentist model (e.g., with 'expected marginal means' or post-hoc tests) using the estimated effect $\hat{\beta}$, 95% confidence interval, and a p-value (equivalent to one minus p_d).

These values allow us to understand the size of the effect and our confidence in its predicted direction. We use the heuristic that there is strong evidence an effect exists if the 95% CredI does not include 0, and weak evidence if the 95% CredI includes 0 but $p_d > 0.95$ (Nicenboim and Vasishth, 2016). (Note that this is distinct from effect size, which can be assessed from the values contained in the CredI.) This is analogous to interpreting whether a 95% confidence interval overlaps 0 and whether $p < 0.05$ for a frequentist model.

We report a variety of z below to understand the model's predictions; Tables III–IV shows the posterior summary measures for each z. Our description of each z is fairly high-level (for example, we don't describe averaging over predictors which aren't of interest); see Supplemental Information III.C for details.

C. Model interpretation

The model's predictions for $P(X)$, $P(XY)$, and $P(\text{trochee})$ as `Louder` and `Longer` are varied, for each `TaskOrder`, are shown in Supplemental Information Fig. 1. This plot looks qualitatively similar to Fig. 1, reflecting that the model captures the basic empirical pattern of prominence and grouping (and hence foot) decisions.

There are no clear qualitative differences by `TaskOrder` for the prominence or grouping decisions (panels in the top row look roughly similar, as do panels in the middle row), although there are quantitative differences we will unpack below.

TABLE III. Posterior summaries of effects of interest for the prominence and grouping decisions ($P(X)$, $P(XY)$), and the resulting foot decision (P(trochee)).

Decision	Effect	Estimate	95% CredI	p_d
$P(X)$	`Louder` slope	0.18	$[0.17, 0.2]$	1
	`Longer` slope	0.14	$[0.12, 0.16]$	1
	`SyllableOrder` = YX	-0.01	$[-0.04, 0.02]$	0.74
$P(XY)$	`Louder` slope	0.15	$[0.14, 0.17]$	1
	`Longer` slope	-0.16	$[-0.19, -0.14]$	1
	`SyllableOrder` = YX	-0.37	$[-0.41, -0.33]$	1
P(trochee)	`TaskOrder`: PFirst − 4Choices	0.14	$[0.08, 0.19]$	1
	`TaskOrder`: PFirst − GFirst	0.12	$[0.06, 0.18]$	1
	`SyllableOrder` = YX	-0.02	$[-0.05, 0.01]$	0.85

The foot decision depends on `TaskOrder`: trochee responses are more likely in the ProminenceFirst order than other orders (bottom-right vs. other bottom-row panels). The model confirms that P(trochee) is higher for the ProminenceFirst order than for other orders (vs. FourChoices: $\hat{z} = 0.14$, 95% CredI $= [0.08, 0.19]$, $p_d = 1$; vs. GroupingFirst: $\hat{z} = 0.12$, 95% CredI $= [0.06, 0.18]$, $p_d = 1$).

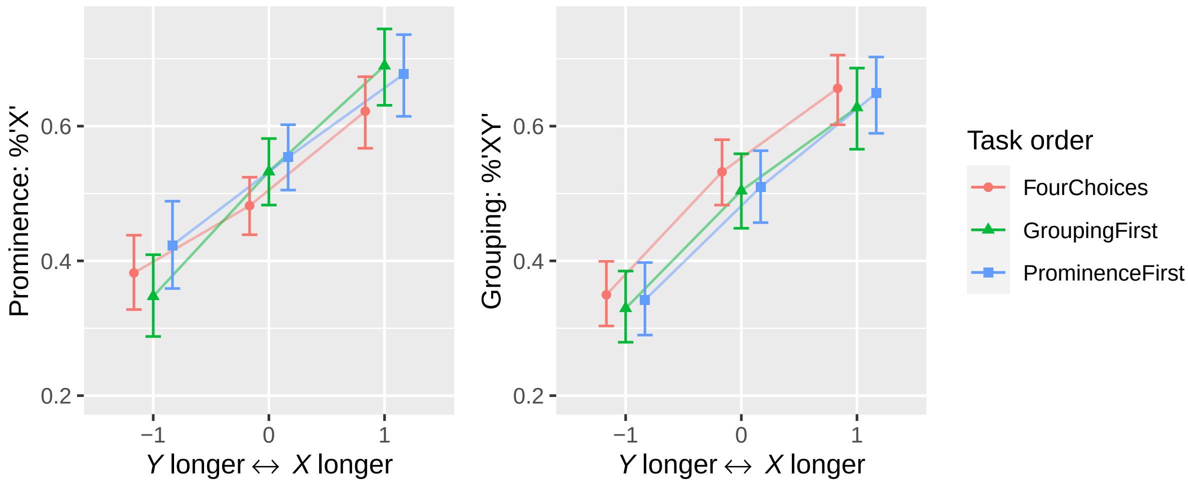

FIG. 8. Predicted effects of duration and intensity on the prominence and grouping decisions, for each task order (marginalizing over participant, syllable order). Dots/errorbars are medians/95% credible intervals of of predicted effects (i.e., the expected values of the posterior predictive distribution).

a. Acoustic cues: effects on each decision. We first examine the model's predicted effects of the acoustic cues, duration and intensity, on the prominence and grouping decisions: Figure 8 shows model predictions for $P(X)$ and $P(XY)$ for each task order.

The qualitative effects are the same across all task orders, so we just give the results averaging across `TaskOrder`, which corresponds to the overall slope for each panel of the figure: higher duration makes it more likely that listeners hear a syllable as final and stressed (`Longer` slope, $P(X)$: $\hat{z} = 0.14$, 95% CredI $= [0.12, 0.16]$, $p_d = 1$; $P(XY)$: $\hat{z} = -0.16$, 95% CredI $= [-0.19, -0.14]$, $p_d = 1$), and higher intensity makes it more likely that participants hear a syllable as initial and stressed (`Louder` slope, $P(X)$: $\hat{z} = 0.18$, 95% CredI $= [0.17, 0.2]$, $p_d = 1$; $P(XY)$: $\hat{z} = 0.15$, 95% CredI $= [0.14, 0.17]$, $p_d = 1$).

There was also strong evidence for an effect of underlying order on the grouping decision (not plotted here): the probability of grouping = XY ($P(XY)$) is lower when `SyllableOrder` is YX ($\hat{z} = -0.37$, 95% CredI = $[-0.41, -0.33]$, $p_d = 1$). This makes sense if listeners tend to hear the first syllable that emerges from the noise as the initial syllable of a word. In other words, the white noise and ramp at the beginning of the sound did not completely remove the order bias. There is no evidence for an analogous effect for the prominence or foot decision (Table III: `SyllableOrder` rows).

b. Mutual influence of decisions. We also examined the model's prediction of how the prominence and grouping decisions are affected by *conditioning* on the other decision. This can intuitively be thought of as examining how the two decisions influence each other (though the model is agnostic on the causal direction of any relationship).

We do this by predicting $P(XY)$ conditional on `Prominence` = X or Y (two probabilities), and $P(X)$ when `Grouping` = YX or XY (two probabilities), which can be computed in terms of levels of `PG` using Bayes' rule. For example,

$$\begin{aligned} P(\texttt{Grouping} = \text{XY} \,|\, \texttt{Prominence} = \text{X}) &= \frac{P(\texttt{Grouping} = \text{XY}, \texttt{Prominence} = \text{X})}{P(\texttt{Prominence} = \text{X})} \\ &= \frac{P(\texttt{PG} = \text{X.XY})}{P(\texttt{PG} = \text{X.YX}) + P(\texttt{PG} = \text{X.XY})} \end{aligned}$$

We write these four conditional probabilities as $P(XY \,|\, X)$, $P(XY \,|\, Y)$, $P(X \,|\, XY)$, and $P(X \,|\, YX)$.

Figure 9 shows these conditional probabilities, as a function of `Louder` and `Longer`, for each `TaskOrder`.

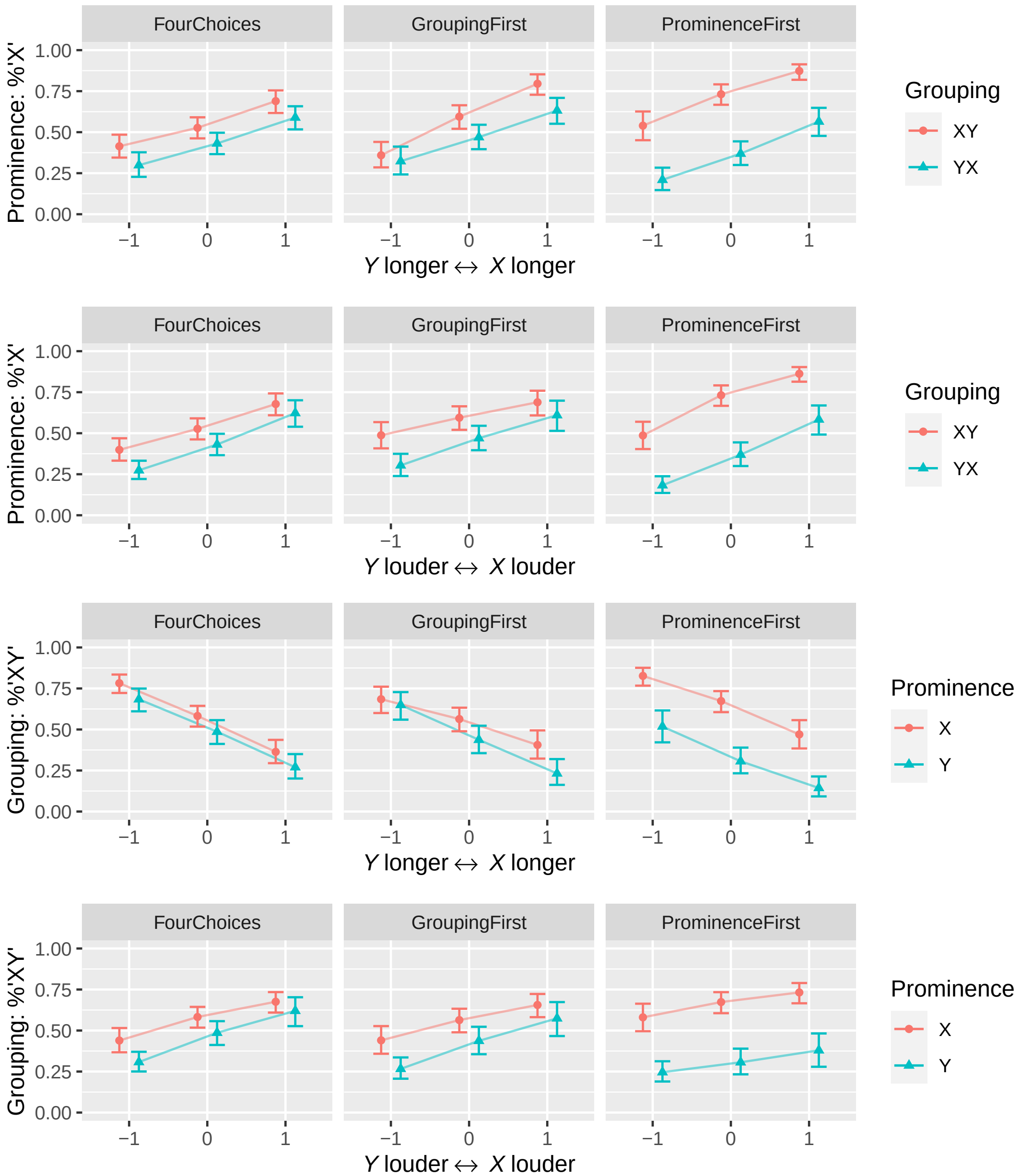

FIG. 9. Predicted effects of duration and intensity on the prominence and grouping decisions, for each task order, conditioning on the other decision (marginalizing over participant, syllable order). Dots/errorbars as in Figure 8.

We can examine the degree of mutual influence by considering just the plots for the `Prominence` decision (top rows), considering first the 'average' influence (vertical difference between red and blue lines in the same panel), i.e., $P(X \mid XY) - P(X \mid YX)$, and then the influence on the slope of the effects of acoustic cues. The degree of influence of the `Prominence` decision on the `Grouping` decision in this model turns out to be identical (in log-odds), by the 'independence of irrelevant alternatives' assumption of the multinomial model used.[13] For example, $P(X \mid XY) - P(X \mid YX)$ is the same as $P(XY \mid X) - P(XY \mid Y)$ (in log-odds). Thus, we can discuss *mutual influence* of the decisions, without specifying which one is influencing the other. For this reason, we only show posterior summaries for how `Prominence` is influenced by `Grouping` in Table IV. Because model prediction plots are in probability, the influence of each decision on the other will not look identical (e.g., vertical distance between red and blue lines in the top-left and bottom-left panels of Fig. 10), so we show plots for both conditional decisions, for clarity.

For average influence: `Grouping` influences the `Prominence` decision (red > blue) regardless of task order (Table IV: rows 1–3), and the degree of influence is larger in the Prominence-first task order than other task orders (rows 4–5).

Turning to the effects of acoustic cues: there are only small differences in the effects of `Louder` and `Longer` on a given decision, conditioning on the other decision. The largest differences are for the `Longer` and `Louder` effects for the GroupingFirst task order, where the grouping and prominence decisions influence each other slightly more as duration increases/intensity decreases, respectively. These are also the only cases where there is strong evidence a difference exists (95% CredI doesn't overlap 0 in Table IV: rows 6–11); there is also weak evidence a difference exists for the `Louder` effect for the FourChoices order.

TABLE IV. Posterior summaries of effects of interest for mutual influence of the prominence and grouping decisions, which can be interpreted either as $P(X|XY) - P(X|YX)$ or $P(XY|X) - P(XY|Y)$.

Effect	TaskOrder	Estimate	95% CredI	p_d
Intercept	FourChoices	0.1	$[0.04, 0.16]$	1
	GroupingFirst	0.11	$[0.05, 0.18]$	1
	ProminenceFirst	0.3	$[0.24, 0.36]$	1
	PFirst - 4Choices	0.2	$[0.11, 0.28]$	1
	PFirst - GFirst	0.18	$[0.1, 0.27]$	1
Longer	FourChoices	0.04	$[-0.16, 0.23]$	0.62
	GroupingFirst	-0.33	$[-0.56, -0.1]$	0.99
	ProminenceFirst	-0.09	$[-0.33, 0.15]$	0.72
Louder	FourChoices	0.16	$[-0.02, 0.34]$	0.93
	GroupingFirst	0.22	$[0.01, 0.42]$	0.96
	ProminenceFirst	-0.03	$[-0.24, 0.19]$	0.58

Thus, there was strong evidence that the decisions influence each other, resulting in a trochee bias. This is especially true in the Prominence-first task order. There is less evidence that the acoustic effects for one decision are modulated by the other decision.

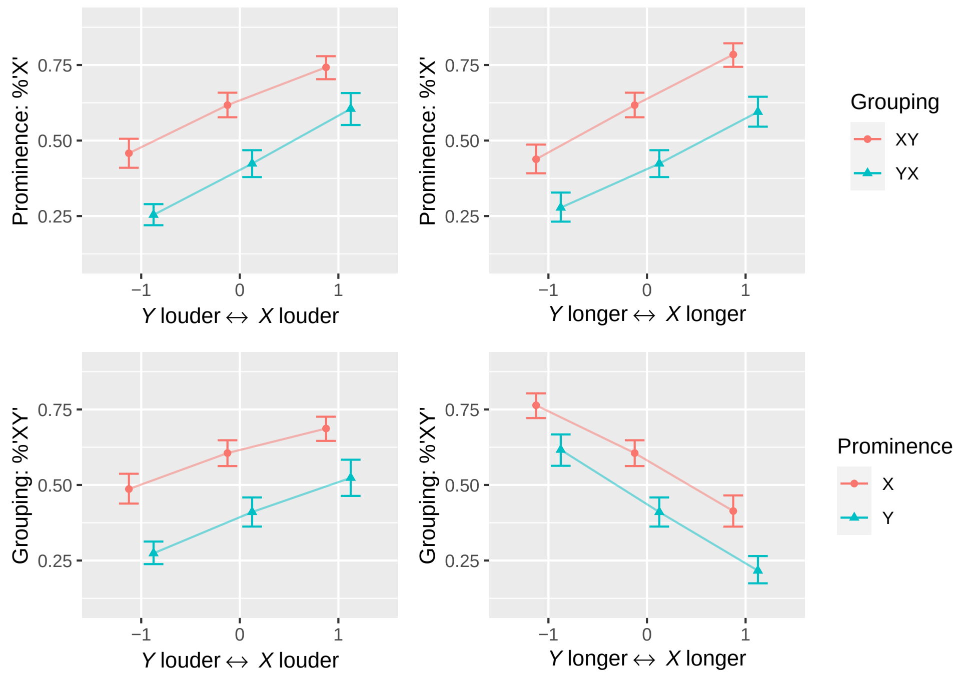

FIG. 10. Predicted effects of duration and intensity on the prominence decision conditioned on grouping decision (top row), and vice versa (bottom row), marginalizing over task order, syllable order, and participant. Dots/errorbars as in Figure 8.

706 The overall pattern is clearer in a model-prediction plot averaging across task orders (Figure 10): the direction of the effects of duration and intensity are consistent irrespective of the other the other decision.

709 The pattern is strikingly different for the foot decision: Figure 11 shows model predictions of P(trochee) as a function of each acoustic cue, conditioned on the `Prominence` and `Grouping` decisions. Here, the effects of duration and intensity for each decision are reversed depending on the other decision, as already observed in the empirical plots (Fig. 6).

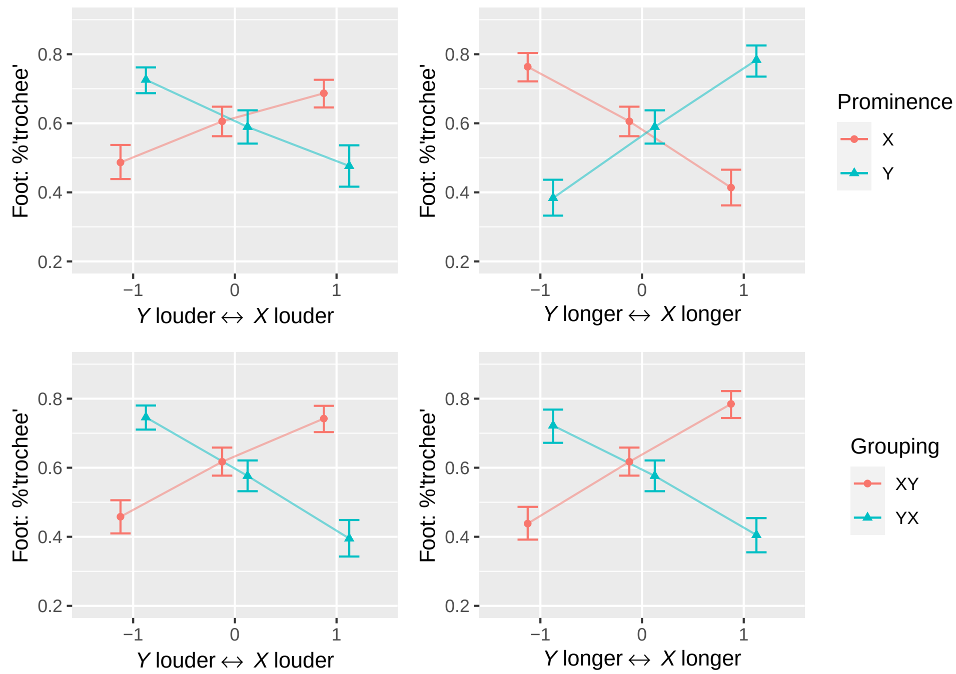

FIG. 11. Predicted effects of duration and intensity on the foot decision conditioned on grouping decision (top row), and prominence decision (bottom row), marginalizing over task order, syllable order, and participant. Dots/errorbars as in Figure 8.

IV. DISCUSSION

The experiment showed strong evidence that in English, an increase in syllable duration makes a syllable more likely to be perceived as stressed and final within a word, and that an increase in intensity makes a syllable more likely to be perceived as stressed and initial within a word. These findings replicate with real-word sequences earlier results based on nonce-word sequences and tone sequences reported by Wagner (2022), which supports the idea that these results may generalize to the perception of more natural speech.

The results also replicated that, contrary to assumptions in the literature on the ITL, intensity and duration are poor cues for the distinction between iambs and trochees. We saw that cues for a particular foot type are inconsistent and depend on the decisions about the other dimensions. In this sense, the quest for cues for iambs vs. trochaic perception is futile: There are consistent cues to foot type only when we condition on more basic percepts about either prominence or grouping. This also illustrates why exploring the effects of intensity and duration on speech segmentation by looking at forced choices between foot types, as is often done in studies on the ITL, by design will lead to poor perceptual differentiation of the stimuli. The poor relation between acoustic cues and foot type is exactly as expected if listeners interpret excess intensity and duration as cues for prominence and grouping.

Grouping and prominence are in principle orthogonal to each other. But the results here show that the perception of these two dimensions is not entirely orthogonal. In line with many prior studies on the ITL, we found a trochee bias in our responses, and moreover showed that this trochee bias is modulated by task order. We discuss the potential sources of this trochee bias and the modulation by task in the general discussion.

All three task manipulations go beyond earlier studies on the ITL which either only asked, in some form or other, a foot choice (e.g., "Did you hear trochees or iambs?") (e.g., Bell, 1977; Bhatara *et al.*, 2013; Hay and Diehl, 2007; Iversen *et al.*, 2008; Rice, 1992; Vos, 1977), or a grouping choice (Abboub *et al.*, 2016; Bhatara *et al.*, 2016; Bion *et al.*, 2011; Crowhurst, 2016; Crowhurst and Teodocio Olivares, 2014; Hay and Saffran, 2012; Molnar *et al.*, 2014; Yoshida *et al.*, 2010). As discussed, these binary forced choice tasks only narrow down the possibilities to two out of the four options, and thus under-determine the exact percept a listener experiences, whereas establishing the exact percept is essential if we want to understand the relation between the prominence and grouping choices.

Earlier studies using the foot task, however, only varied one acoustic variable at a time, and the assumption may have been that listeners will always hear the longer/louder syllable as more prominent, in which case we would know the exact percept after all. This assumption turns out to be empirically incorrect, however: Among the stimuli in which exactly one syllable was manipulated, the longer syllable was heard as prominent only 64% of the time, and the louder syllable only 65% of the time. This makes sense under the hypothesis here, since instead of attributing higher intensity/duration to stress, listeners can attribute them to initiality/finality.This replicates similar findings Wagner (2022).

Our three tasks share with earlier studies that listeners had no way to report whether their percept changed over the course of listening to it. It seems plausible that this could happen, since based on our hypothesis, stimuli should often have more than one stable interpretation consistent with the cues. For example, an increase in duration in one syllable could be interpreted as a cue that the syllable is final, leaving open whether it is prominent or not; or it could be interpreted as a cue to prominence, leaving open whether it is initial or final within the group. According to our hypothesis, this manipulation only precludes the parse in which the syllable is initial and unstressed, but is compatible with the other three parses. Similarly, if one syllable is longer and the other loud, the cues are a consistent signal for grouping, but leave open which syllable is stressed. If the same syllable is longer and louder, the cues send a consistent signal for prominence, but inconsistent cues for grouping.

The stimuli are thus similar to Neckar cubes in the visual domain, which have multiple stable interpretations, and it would not be unexpected if a listener's percept varied over time, vacillating between the multiple possible interpretations of the cues (cf. Wagner, 2022; Warren and Gregory, 1958). However, even if a participant had multiple different interpretations over the course of a trial, their decisions will still reveal *some* consistent interpretation they gave to the signal.

Overall, our results showed strong evidence for the direction of the intensity and duration effects on the grouping and prominence decisions, supporting our reinterpretation of the ITL. They did not, however, show clear evidence for Bolton's ITL generalization itself—that manipulating duration alone results in a greater likelihood of perceiving iambs and manipulating intensity alone results in a greater likelihood of perceiving trochees. This raises the question whether our current hypothesis can actually provide an alternative explanation for earlier results. We did not find a clear ITL pattern, in that only manipulating duration only slightly shifted responses toward iambic, and only manipulating intensity did not seem to increase the trochee rate by much. Given the model predictions, this is not surprising, but the expectation is that we *would have* found the ITL pattern more clearly with more extreme manipulations of the individual cues—just like earlier ITL studies, which indeed manipulated intensity and duration in rather extreme ways. We tested this prediction with a second experiment which involved such more extreme manipulations, as a way to ensure that our methods tap into the same generalization as earlier studies on the ITL.

V. EXPERIMENT 2

Experiment 2 was identical to the Four-way task in the previous experiment, except that we used more extreme duration and intensity manipulations. We varied the intensity difference between 0 and 12dB above the baseline, and duration between 0 and 160ms above the baseline. Note that both for intensity and for duration, the differences between the manipulated and non-manipulated syllables far exceed differences due to grouping or prominence observed in actual productions (e.g., those reported in Wagner, 2022). A total of 75 participants participated in the experiment, but only the responses of 61 of them were used in the analysis. We excluded 3 participants who didn't finish the task, 1 participant whose first language was not English, and 10 participants who failed the headphone screening test.

A. Results

Fig. 12 illustrates the results. When looking at the foot decision, there is now a much clearer ITL pattern: For extreme manipulations of duration (the outermost cells in the middle row of the foot decision heat map), listeners were more likely to perceive iambs than trochees, as evidenced by the red shading of the cells. And for extreme manipulations of intensity (the outermost cells in the middle column of the foot decision heat map), listeners were more likely to perceive trochees, as evidenced by the blue shading of the cells.

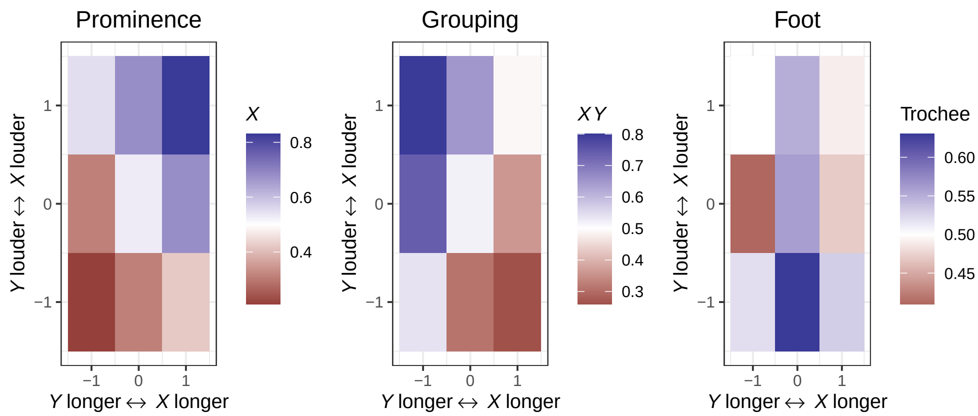

FIG. 12. Prominence, grouping, and foot decisions for the second experiment which involved more extreme manipulations.

B. Discussion

The plots of our follow-up experiment show Bolton's basic ITL pattern: If one manipulates the duration of every other syllable, listeners are more likely to perceive iambs. If one manipulates the intensity of every other syllable, listeners are more or equally likely to perceive trochees.

Hay and Diehl (2007) (among others) suggested that duration and intensity are cues for iambs or trochees in a more direct sense, and posited that listeners parse the signal for foot type, following earlier ideas (e.g., Hayes, 1995). However, the fact that the ITL pattern only emerges when manipulating duration or intensity to an extreme extent, unlikely to occur in natural productions, casts doubt on the idea that foot perception plays a primary role in parsing the signal.

We already saw that putting feet at the core of a principle of speech parsing is also incompatible with the fact that overall, intensity and duration do not reliably cue foot type, and actually have inconsistent effects on foot choice unless one conditions on other perceptual decisions.

Experiment 2 also provides further evidence that the foot task does not reliably establish the percept of the listener even for stimuli in which only one cue was varied, and even for more extreme manipulations like the ones used here: Among the stimuli in this experiment in which exactly one syllable was manipulated, the longer syllable was heard as prominent only 64% of the time, and the louder syllable only 65% of the time.

VI. GENERAL DISCUSSION

The results reported in this paper are compatible with Bolton's classic ITL pattern as an empirical generalization about certain kinds of stimuli with extreme duration and intensity manipulations, but they cast this pattern in a new light. There are two main findings that we discuss in turn.

A. The Iambic-Trochaic Law is not about iambs or trochees

Our data provides new evidence that the ITL pattern is not really about perceptual cues to iambs or trochees as such, it is an indirect result of listeners parsing the acoustic stream along the dimensions of grouping and prominence.

The ITL emerges when one manipulates either intensity or duration in an extreme way. If a manipulation in duration/intensity is too extreme to be attributed to one of these two perceptual dimensions alone, listeners will conclude that a syllable must be final and stressed (in the case of duration) or initial and stressed (in the case of intensity). Looking at a broader range of stimuli, however, shows that there is a very consistent relation between duration/intensity and both grouping and prominence, but there is no such consistent relation with foot type, even if some earlier studies were misled into thinking that there are consistent cues to foot type based on responses to extreme cases.

Even when the task does not, by virture of the questions asked, make particular dimensions salient, listeners treat intensity and duration as cues to prominence and grouping rather than foot type. This suggests that foot perception is derivative of the cognitively more basic decisions of grouping and prominence, and the notions of trochee and iamb may not play any direct role in speech parsing. While prominence and grouping decisions of course settle the foot type, there are no systematic cues that distinguish different foot types.[14]

Rethinking Bolton's 'general principle' and the later proposals about a psycho-physical 'Iambic-Trochaic Law' does not invalidate earlier results that were obtained with experiments based on foot tasks. However, it shows why earlier experiments might have had resulted in more interesting findings if grouping and prominence (or at least one of them) had also been considered in addition to foot type. It is necessary to consider the full range of possible percepts, and tease apart the dimensions of grouping and prominence.

An explanation of Bolton's principle without direct reference to foot type may be desirable independently of the empirical problems with this account discussed here. It is not obvious that the notion of 'trochee' and 'iamb' should be considered cognitive or linguistic primitives. Throughout the paper, we have used the terms 'trochee' and 'iamb' descriptively to refer to a bisyllabic word with initial stress or final stress respectively. The notions of trochee and iambs played no role, however, in the explanation of the data. This seems desirable, since it has been shown that similar effects as the ITL are observed if every third syllable is made louder or longer (Bell, 1977; Vos, 1977). In the former case, listeners tend to hear groups of three sounds with initial prominence, in the latter case, listeners tend to hear groups of three sounds with final prominence. This has not led to a proposal of a dactylic/anapestic law, and nor is one called for under the perspective taken here.

Some theories of word stress posit more technical notions of feet, making different assumptions about what constitutes a 'trochee' or an 'iamb.' Hayes (1995), for example, proposed a theory of metrical theory that treats the English word *out* and *look* by themselves as constituting a trochaic foot, since they have two morae, or units of eight. Our results here suggest that Bolton's generalization is not sensitive to this technical notion of foot proposed by Hayes and others, but rather is about the simpler notion about whether stress within a group is initial or final, irrespective of the weight of the syllables involved. In that sense Hayes's definition of foot type turns out not to be relevant for the ITL. This casts some doubt on Hayes's use of the ITL as an argument in support of the proposed universal foot inventory. But we will not explore these questions further here, since ultimately Hayes (1995) wanted to explain the typology of word stress systems, and not the ITL, and we do not address this typology here. See Gordon (2007) for a view of word stress and syllable weight that does not involve a phonological category of 'phonological foot'.

Another reason why an explanation that is not tied to the notion of word-level feet seems desirable is that the cue distribution at higher levels of prosodic organization above the word, in particular the cues to prosodic phrasing and the cues to sentential prominence-relations (as influenced, for example, by focus and contrast), appear to be exactly parallel to those at the word level reported by Wagner and McAuliffe (2019): When intensity and duration correlate they encode phrasal prominence; when they anti-correlate, they encode phrasal grouping. The regularities observed in the ITL pattern are thus likely not about word segmentation and word stress per se, where linguistic foot structure is typically assumed to play a role. Rather than pointing to word-level foot structure, the signature effects of duration and intensity across domains reveal something much more general about how we parse speech.

In fact, several of our 'words' were, from a syntactic point of view, phrases. The phrasal verbs *set up*, *set in*, and *turn over* are unlike words in that they can be separated from each other by other words and phrases (as in *turn something over*).[15] This distinction between phrases and words did not appear to matter for our experiments.

We did not discuss the potential causes of the observed cue distribution. One cause may be perceptual Gestalt principles such as the principle of proximity (Wertheimer, 1923). The duration effect on grouping observed here, for example, may occur because sounds whose attack points are closer to each other tend to be grouped together to the exclusion of sounds whose attack point is more distant (Lerdahl and Jackendoff, 1983). A reset of intensity after a period of downdrift may induce grouping due the Gestalt principle of common fate. Finally, figure-ground principles may lead to a percept of prominence when acoustic energy of a sound is increased by greater duration or intensity. These Gestalt principles are not specific to language. Wagner (2022) showed evidence the sequences of sinusoidal tones are also parsed for grouping and prominence, and acoustic cues are interpreted in parallel ways (see Heffner and Slevc, 2015; Katz, 2022, for reviews of parallels in prosodic structuring between music and speech).

But the source of the cue distribution in speech and music may also reflect *production* constraints. For example, the reason why in speech, intensity gradually decreases ('downdrift') rather than increases within a constituent is likely a reflex of the drop in subglottal pressure during exhalation, an effect that may have been 'grammaticalized' as a cue to grouping in some languages. And final lengthening, which just like intensity downdrift recurs across languages, although to different degrees (Paschen *et al.*, 2022), may be grounded in the slow-down of movement observed at the end of motor plans. Similar 'biological codes' have been observed in the domain of pitch (Gussenhoven, 2004).

B. Prominence-first leads to a greater trochee bias

The second central finding of this study is that the trochee bias (observed here and in earlier studies) is stronger when participants are first asked about their prominence percept, and then about their grouping percept, or if they are given a choice between four options. To better understand why, we need to understand what the trochee bias might be due to.

One possible source of a trochee bias is that listeners apply their lexical knowledge of English to parse the speech stream. Cutler and Carter (1987) proposed that listeners my take stress as a cue for word beginnings because about 70% of bisyllabic words of English have trochaic stress. Studies since then have found more evidence for the inference that stressed syllables must be initial (Cutler and Butterfield, 1992; Cutler and Norris, 1988;

Mattys *et al.*, 1999), although this does not explain why Wagner *et al.* (2021) found a similar bias in French.

A second possibility is that trochee bias results from a general cognitive bias to hear the first element of a group as prominent. This bias has been observed in earlier psycho-acoustic studies. In equitone sequences which are equally spaced, listeners who hear them as grouped tend to hear initial prominence (Povel and Okkerman, 1981). The reason for this might be the same as the source of the downbeat in music. In music (at least in Western music), the first beat in a bar is a position which is perceived as 'accented' (cf. Patel, 2007). This is the most likely place for a tap if one taps along to music, and in music performance a note in this position is often played with greater emphasis. This 'downbeat effect' is a property of the overall rhythmic pattern we impose, and not (necessarily) of the acoustic properties of the musical note in the downbeat position. A downbeat can still be felt if no note at all is played at all in that position.

Neither of these two possible sources seems to explain why the bias would be modulated by task order, such that asking for prominence first increases it. We had not predicted an effect in this particular direction, and we can only speculate as to why it exists.

One interpretation offers itself once we become more specific about how exactly a listener may end up concluding that they are listening to trochees. A trochee bias could arise because listeners hear a syllable as prominent, and then, in the absence of clear cues to grouping, infer that the syllable must be initial within a group. Alternatively, it could arise if a listener hears a group, and then, in the absence of clear cues to prominence, infer that the first sound in the group must be prominent.

If ordering the prominence task first has the effect that the prominence decision is more likely to be made before the grouping decision, then the increase in the trochee bias in this order can be taken as evidence that listeners tend to draw the former inference, *prominent* → *initial*, but not the latter, *initial* → *prominent*.

There is independent evidence that supports the presence of a *prominence* → *initial* inference. Lahiri and Plank (2010) review a long tradition of research that found that the stretch from one prominence to the beginning of the next prominence is treated by listeners and/or speakers as a unit, even if it mismatches the grammatical structure of the utterance (i.e., irrespective of word and constituent structure). For the example, the phrase *towards the land* can prosodically be grouped as *(towards the) (land)* Lahiri and Plank (2010, 372). These units have alternately been called 'cadences' (Steele, 1775, 11), 'sound groups' (Sweet, 1876, 473), or 'Sprechtakte' (Sievers, 1901, 234). Abercrombie (1964, 10) uses the term 'foot' (sometimes referred to now as 'Abercrombian foot'): "A foot, in this usage, may be defined as the space in time from the incidence of one stress-pulse up to, but not including, the next stress-pulse."

Lahiri and Plank (2010) and Bögel (2020) present new evidence in favor of this prosodic unit, both from production studies and from historical change. An inference of the form *prominent* → *initial* could derive this parse when perceiving speech. While we have no specific hypothesis about why this perceptual bias should exist, our perception data suggest that it does, and furthermore that there is no, or at least only a weaker, counterpart of the form *inital* → *prominent*. Lahiri and Plank (2010) actually locate the source of this prosodic parsing preference in language production, where the size of production planning units is sensitive to prominence.

Note that this way of thinking about the trochee bias is very different compared to the lexical statistics hypothesis, which grounds the bias in independent knowledge about independently motivated grammatical structures with initial prominence (i.e., the actual words in the lexicon). The *prominent* $\rightarrow$ *initial* bias here leads to parses that *violate* the predominant independently motivated structure.

If listeners have a *prominent* $\rightarrow$ *initial* bias, but not the converse, then observing a stronger effect when the prominence question is asked first is expected, since then prominence information is available (or at least made salient) earlier relative to grouping information. The fact that the Four-way task and the Grouping-first task are very similar to each other would then suggest that typically (unless asked about prominence first), listeners first decide grouping, and then prominence.

However, we should mention one caveat. This line of reasoning is based on the premise that task order exerts its effect by modulating which perceptual decision is made first, prominence or grouping. This seems plausible to us, but it is not a given. Further studies are necessary to understand why exactly asking about prominence first has the effect that it does.

C. The ITL and speech rhythm

The results of this study can inform the broader question of whether there are universal aspects to how we parse speech. Rather than looking for cross-linguistically valid cues for feet, we should establish whether there are cross-linguistically robust cues for prominence and grouping, parallel to the Gestalt principles involved in organizing the sensory input observed in other cognitive domains. Wagner *et al.* (2021) explore this question to some extent, and show that languages appear to vary in how intensity and duration are interpreted as cues to grouping and prominence, but also that there is a cross-linguistic consistency for duration as a cue for prominence, and intensity as a cue for grouping. More production and perception studies looking at a greater number of cues and languages will be necessary.

The results can also inform the debate about whether languages fall into different rhythmic classes, and if so how these might be defined. Dauer (1983) convincingly showed that it is useful to 'decompose' the notion of rhythm into different factors. Some quantitative measures of the 'rhythm' of a language (Low *et al.*, 2000; Ramus *et al.*, 1999) arguably reflect phonotactic and phonemic differences between languages (Arvaniti, 2012). Grouping and prominence, on the other hand, capture core aspects of what we intuitively call 'rhythm' in daily speech, and Wagner *et al.* (2021) show that the way languages vary in how they use intensity and duration to convey grouping and prominence can be used to quantify rhythmic differences. Crucially, our finding that perceptual decisions about grouping and prominence show some degree of mutual dependence, converging on a trochee bias, show that they are really connected, validating the idea that the two dimensions integrate in some sense to an overall percept of 'rhythm.' The cognitive or psychoacoustic underpinnings of this trochee bias remain to be understood.

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[1]The unusually great intensity difference in Hay and Diehl (2007) has the effect for some listeners that the softer syllable of the *gaga* sequence sounds like an echo of the louder syllable, an effect parallel to auditory streaming effects (cf. Bregman and Campbell, 1971).

[2]Though see Turk and Shattuck-Hufnagel (2000) for an experiment which failed to find word-final lengthening.

[3]`https://github.com/prosodylab/prosodylab-experimenter`

[4]See the supplementary materials at https://www.scitation.org/doi/suppl/10.1121/10.0017170 a for additional information about the task, the participants, and the statistical models. See the OSF repository at https://osf.io/bsfvu/ for the full set of acoustic stimuli, as well as the data and code for the analysis. The OSF respository also includes the preregistration, which can be directly viewed accessed here: https://osf.io/58yk6.

[5]`https://aws.amazon.com/polly/`

[6]See Supplemental Information for a sample set of instructions and the full list of definitions.

[7]The mic check also ensured they could participate in the short production experiment that followed the perception experiment. In the production study, results of which are not reported here, participants were recorded on sentences containing each of the 16 words used in the perception study.

[8]The bivariate Bayesian model reported in in the Supplementary Information of Wagner 2022 has related issues, which we will not discuss here.

[9]The actual formula is slightly more complicated because of the monotonic effects and how `brms` codes multinomial models; see Supplemental Information Sec. III.

[10]For example, we would not be able to use monotonic effects, or fully model by-participant variability.

[11]Given the large sample size, we expect all results reported below would be very similar for different priors (including for 'flat' priors, which would be equivalent to a frequentist model).

[12]A technical note is that we compute z in probabilities, rather than log-odds, to ease interpretation.

[13]Whether this assumption holds is an empirical question, but for this particular dataset, Figures 3–4 suggest it is not grossly violated.

[14]We did not run an experiment on the stimuli here asking directly for a foot decision (in addition to a forced choice of either grouping or prominence). See Wagner (2022) for evidence that even then, listeners interpret the cues as cues for pominence and grouping, while there are no systemtic cues for foot type as such.

[15]It is less obvious whether *look out* is phrasal in this sense as well, since the two morphemes cannot be easily separated from each other.

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